# Comprehensive System Architecture and Technical Report

*Orbius: Multi-Agent BRD & Living Requirements Studio*

## 1. Executive Summary

**Orbius** is an enterprise-grade, highly scalable multi-modal AI platform designed to reimagine the traditional, static Business Requirements Document (BRD) through a dynamic "Living BRD" framework. Leveraging advanced generative AI (Google Gemini) and a sophisticated multi-agent orchestration architecture, Orbius continuously ingests fragmented, contradictory data from diverse unstructured and structured sources--including emails, PDFs, voice notes, and wireframes.

By executing parallel analysis and identifying conflicting requirements in real-time, the platform automates the synthesis of a cohesive, validated requirement repository. Furthermore, Orbius bridges the critical gap between product management and software engineering by automatically generating executable test suites and interactive UI prototypes from the synthesized requirements.

## 2. Problem Statement & Solution Architecture

### The Challenge of Fragmented Requirements

In modern enterprise product development, project requirements are rarely centralized or consistent. They are dispersed across various mediums: executive emails, recorded voice calls, formal PDFs, financial spreadsheets, and design wireframes. Critically, these sources frequently contradict one another. This systemic fragmentation leads to misaligned expectations, severe scope creep, and expensive late-stage architectural revisions.

### The Orbius Solution

Orbius resolves these inconsistencies by actively ingesting and parsing all multimodal inputs in parallel. Utilizing a specialized AI agent swarm, the platform performs the following operations:

- **Omnichannel Ingestion:** Captures data seamlessly from various file formats and live interactions.
- **Parallel Extraction:** Dedicates specialized agents to extract functional, non-functional, security, and operational requirements.
- **Validation & Resolution:** Deduplicates redundant claims, eliminates ungrounded assumptions, and explicitly flags contradictions for human-in-the-loop resolution.
- **Living BRD Maintenance:** Serves as a dynamic, single source of truth that evolves as new data is introduced into the system.

## 3. Core Features & System Capabilities

> **Real-Time Voice & Multi-Modal Processing:** Native support for ingesting PDF, DOCX, CSV/XLSX, image-based wireframes, and live in-browser voice notes utilizing Web Speech API integration.

> **Parallel Specialist Agent Swarm:** Deploys dedicated AI agents (Text, Vision, Document/Budget, Risk, and Operations) to deconstruct and analyze documents simultaneously, ensuring comprehensive coverage of all project dimensions.

> **Interactive Architectural Diagramming:** Automatically synthesizes and renders system architectures, user flows, and sequence diagrams via Mermaid.js, derived directly from the extracted requirements.

> **Clickable Interactive Prototypes:** Compiles the validated BRD into functional in-browser UI prototypes (e.g., kiosk simulations), allowing stakeholders to visualize the end-product before engineering commences.

> **Automated Gherkin Test Suites & Backlogs:** Auto-generates structured Gherkin test scenarios (Given/When/Then) alongside a test runner simulation, with seamless 1-click exports to Jira (CSV) and Linear (JSON).

> **Live Meeting Mode & Real-Time Conflict Interrupter:** Actively monitors live discussions and dynamically interrupts when spoken statements contradict previously established commitments, such as budget caps or launch targets.

> **Grounded AI Copilot:** Delivers ultra-concise, evidence-grounded answers equipped with clickable citation chips that link directly to the source document chunks, ensuring strict zero-hallucination fidelity.

> **"What-If" Scenario Simulator:** Calculates cascading ripple effects and trade-offs for hypothetical project changes, including budget cuts, scope adjustments, or accelerated timelines.

## 4. System Architecture & Technology Stack

Orbius is constructed upon a robust, decoupled architecture optimized for low-latency processing, resilience, and real-time interaction.

### Frontend Interface

- **Framework:** React 18 and Vite for high-performance, reactive rendering.
- **Visualizations:** Mermaid.js integration for dynamic diagram generation.
- **Interactivity:** Web Speech API for voice ingestion and real-time UI updates powered by Server-Sent Events (SSE).

### Backend Engine & Processing

- **Framework:** FastAPI deployed via Uvicorn (Python 3.11+) ensuring asynchronous, non-blocking I/O operations.
- **Data Validation:** Pydantic v2 for rigorous type checking and schema enforcement across all API boundaries.
- **Processing:** Asynchronous execution model with specialist agents running in parallel threads; progressive requirement delivery via SSE.
- **Data Persistence:** SQLite for efficient, lightweight persistence of session states, tasks, alerts, and user profiles.
- **Ingestion Pipeline:** Utilization of `pypdf`, `openpyxl`, `Pillow`, and `python-docx` for comprehensive, fault-tolerant document parsing.

### AI Layer & Resilience Mechanisms

- **Primary Intelligence:** Advanced integration with the Google Gemini API.
- **Multi-Tier Model Router:** A resilient 3-tier model router with automatic fallback protocols to handle rate limits and service interruptions.
- **100% Offline Capability:** If network connectivity is lost, Orbius autonomously falls back to a deterministic, heuristic-based synthesis engine, ensuring uninterrupted operation in secure or offline environments.

## 5. Data Flow & Processing Pipeline

The data pipeline within Orbius guarantees strict traceability and transparency. Every extracted finding retains its origin `source_id`, exact text span, and calculated confidence score.

1. **Ingestion:** Structured and unstructured data sources are uploaded and parsed into unified `SourceChunks`.
2. **Analysis via Agent Swarm:** The Text, Vision, BVAgi And, and Risk agents process the chunks to emit `Findings`, while the Operations agent identifies and schedules `Tasks` and `Alerts`.
3. **Validation:** A centralized validation engine deduplicates findings, filters low-confidence extractions, and surfaces conflicting statements.
4. **Synthesis & Composition:** The system synthesizes the validated data into structured BRD sections, architectural diagrams, and sequence events.
5. **Action Center & Export:** The platform coordinates human-in-the-loop approvals for external communications and generates executable exports (Jira/Linear backlogs, PDF/Word documents) while empowering the grounded Copilot session.

## 6. Conclusion

Orbius represents a paradigm shift in requirements engineering and project orchestration. By transforming a static, frequently outdated document into a dynamic **Living BRD**, the platform ensures continuous alignment across all stakeholders. The seamless integration of advanced generative AI with practical operational utilities--such as real-time conflict interruption, automated Gherkin generation, and robust Jira exports--elevates Orbius from a mere analytical tool to a comprehensive, scalable hub for enterprise software delivery.